Lab Activity #3 – Fundamental Processing Exercises in Seismic Unix

Here, we do a series of fundamental processing steps using a single Shot gather and a single CMP gather:

- Amplitude correction (Gain)
- Frequency based Filtering (e.g. Band Pass Filters)
- Frequency/wave number (F/K) Filtering (e.g. Fan Filter)
- Velocity Analysis

Part I:

From a Linux machine, open a new terminal window and go to your working directory then

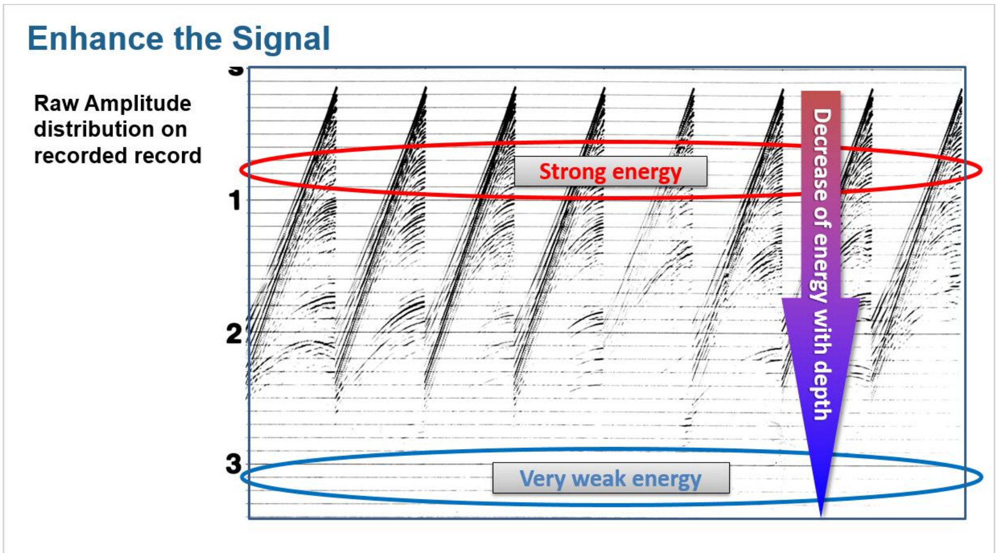
- cp -r ~/Simple_SU_Activities/Data .
- cd Data

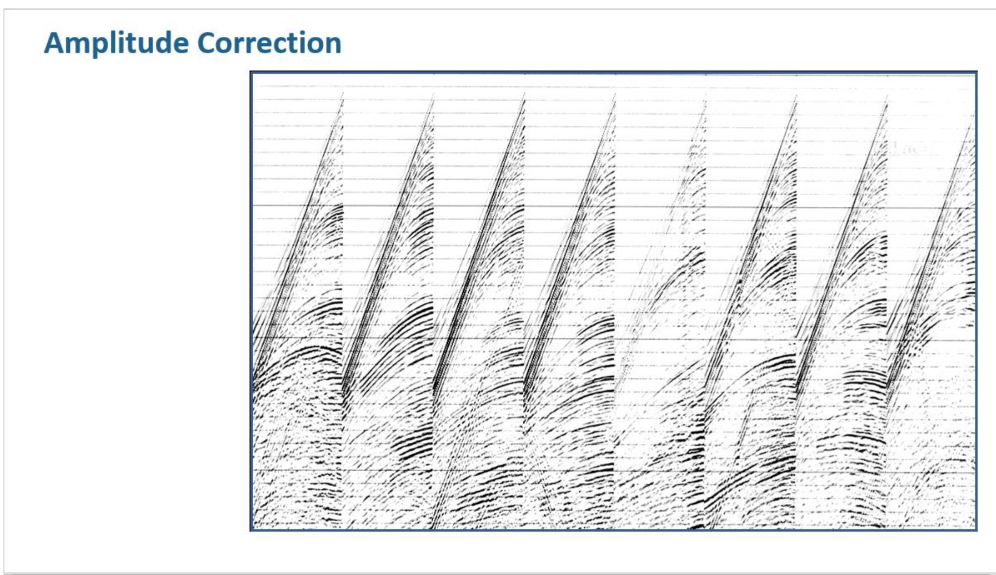
You may need to run the following command: chmod 755 *

Please remember, as you are about to use an SU formatted dataset, to utilize the following su programs: suxwigb, suximage, and surange. Those programs will allow you to learn more about the Gather and the associated header values before running the intended exercise.

Lab Activities:

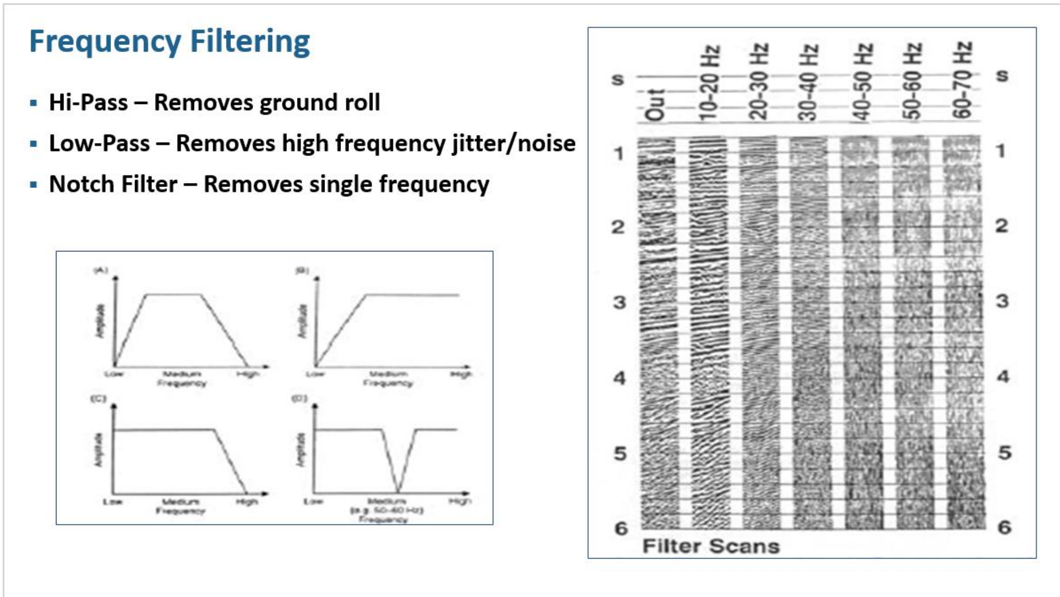
1. use file oz14h4.su to practice variety of amplitude corrections such as AGC and tpow options in sugain program.



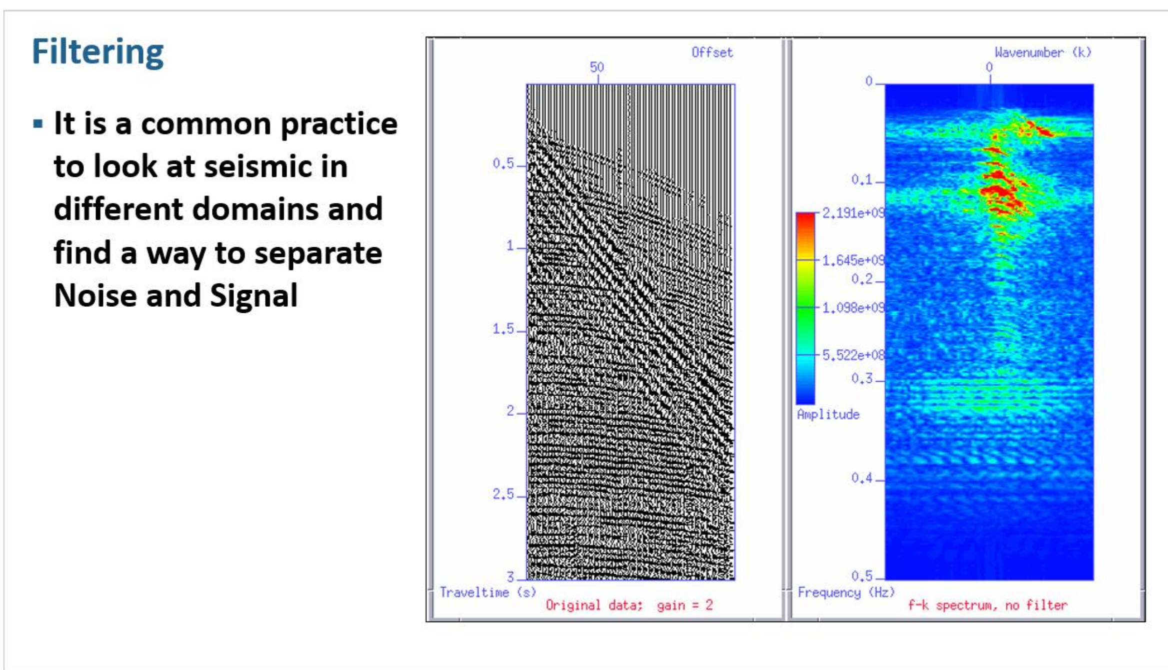


Amplitude correction before and after applying gain.

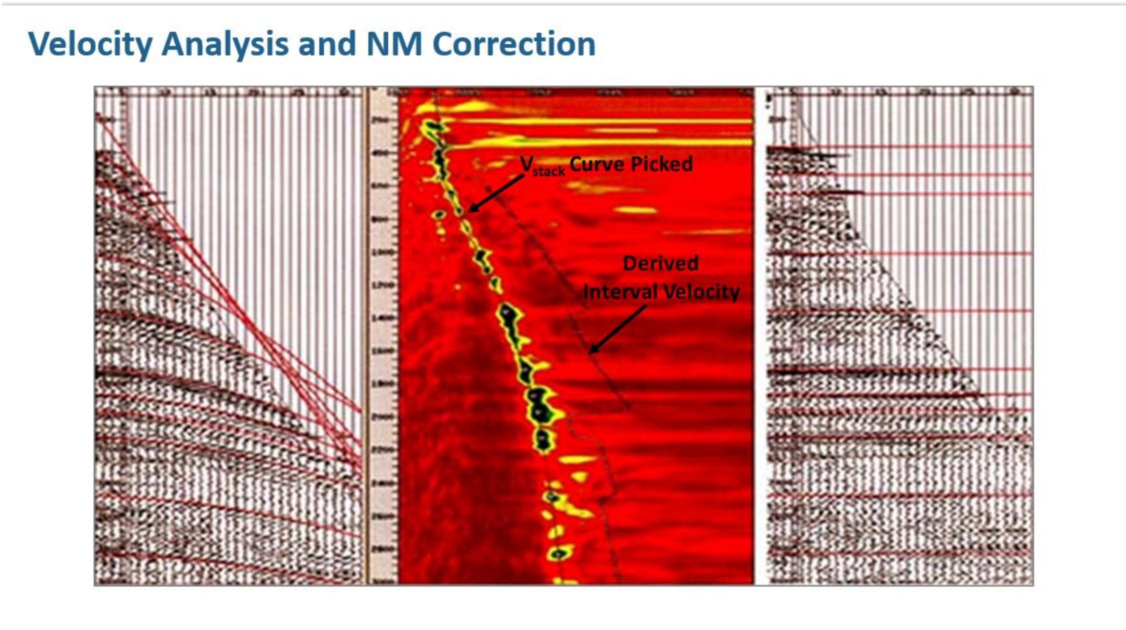
2. use file oz8w.su to practice frequency based filtering such as Band Pass Filter using suspecfx and sufilter programs. You may need to apply gain using sugain such as tpow option.



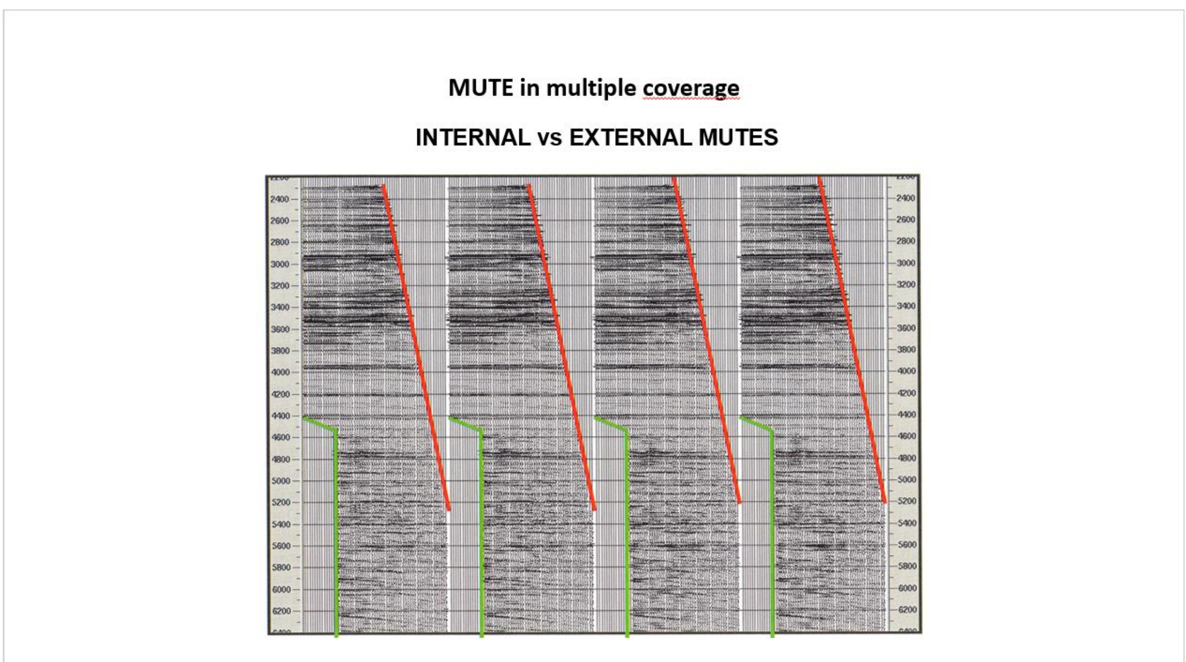
3. use file oz8w.su to practice frequency/wave number based filtering such as a Fan Filter using suspecfk and sudipfilt programs. You may need to apply gain using sugain such as tpow option.



4. use file oz14hv.su to practice conducting proper velocity semblance analysis and then apply your velocity picks to perform NMO correction by using suvelan and sunmo programs; respectively. You may need to apply gain using sugain such as tpow option.



5. use files oz14cmp.su (a CMP gather without NMO correction) and file oz14hvnmo.su (an NMO corrected CMP gather) to practice variety of mute options using sumute program. You may need to apply gain using sugain such as tpow option.



Part II:

Shall you completed your assignment of Part I and you feel you need some additional hints on the exercises, please feel free to access the pre-prepared shell scripts as described below to help you with associated concepts using the same suggested gathers.

But first, and from a Linux machine, open a new terminal window and go to your working directory then

- cp -r /quake/home/dajaniaa/MIT_IAP_Final/Simple_SU_Activities/ .
- cd Simple_SU_Activities
- You may need to run the following command: chmod 755 *.s*

Activity 1 – Amplitude Recovery and Gain:

From a Linux machine, open a new terminal window. At the command prompt, type

- Activity01_igain_dj.sh

Press Enter and follow the instructions in the terminal window.

Activity 2 – F/X Frequency Based Filtering:

From the command prompt, type

- Activity02_ifx_dj.sh

Press Enter and follow the instructions in the terminal window.

Activity 3 – F/K Fan Filtering:

From the command prompt, type

- Activity04_ifk_dj.sh

Press Enter and follow the instructions in the terminal window.

Activity 4 – Velocity Analysis and NMO Correction:

From the command prompt, type

- Activity03_ivelan_dj.sh

Press Enter and follow the instructions in the terminal window.

Please submit a 1-3 page summary of your activities in this lab.